

Cloud Computing

Semester 2026-III

Team Project Assignment

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Team Project Assignment

All teams work within the **Cloud Computing** course framework, designing and building cloud-native distributed systems for popular high-traffic applications. Each team will follow the **Four-Delivery Methodology**:

1. **Delivery 1 — Requirements & Workload Model**
2. **Delivery 2 — Software Architecture**
3. **Delivery 3 — Cloud Architecture & Scalability**
4. **Delivery 4 — Prototype, Resilience & Final Architecture**

Each team has been assigned a unique real-world application with high data volume and traffic characteristics. Teams will design a scalable, resilient cloud architecture appropriate for the application's workload model, from business requirements through cloud deployment and failure scenario analysis.

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Any comment or concern about this document can be sent to Carlos A. Sierra at: *cavir-guezs@udistrital.edu.co*.

Team Project Distribution

Team	Application	Focus Domain	Students
1	Uber / Ride-Sharing	Real-time Geolocation, Matching & Payments	Moreno Barragán Jhonatan David Gaitán Castañeda Luna Valeria Jonathan Felipe Ochoa Silva Ortiz Luna Nicolás Francisco
2	DoorDash / Food Delivery	Order Management, Route Optimization & Notifications	Daniel Felipe Barrera Suárez Carvajal Marín Jaider Camilo Posso Suárez Nelson David García Gaitán Edward Julián Cabrera Barragán Julián David
3	Netflix Streaming Platform	Content Delivery, User Analytics & Recommendations	Gualteros Bustos Santiago Antonio Castro Rivera Nicolás Vargas Medina Andrés Julián Buitrago Peña Erick Santiago
4	Airbnb / Short-Term Rentals	Booking Engine, Search & Inventory Management	Cuellar Cangrejo Daniel David Jiménez Palomino Juan Andrés Bautista Solano Esteban Alexander Reyes Gómez Santiago Ballesteros Molina Daniel Mateo
5	Stripe / Payment Processing	Transaction Management, Fraud Detection & Settlement	Ocampo Gallego Arlo Nicolás Grajales Castillo Juan Diego López Cadena Cristian Santiago Arias Arias Xiomara Salomé
6	Spotify / Music Streaming	Personalization, Recommendation Engine & Media Delivery	Romero Gil Cristian David Arismendi Sánchez Juan Pablo Vargas Castañeda Giovanni Alexander Sánchez Jerez Jan Henrik

Application Workload Characteristics

Each team's assigned application exhibits distinct high-traffic patterns and architectural challenges:

1. Team 1 — Uber (Ride-Sharing):

- **Workload:** 100K+ active rides simultaneously across multiple geographies
- **Characteristics:** Real-time geolocation updates, microsecond matching algorithms, high-frequency database writes
- **Critical Requirements:** Sub-second response times, 99.99% availability, geographic redundancy, real-time event streaming
- **Challenges:** Managing distributed state across availability zones, handling surge pricing computation, coordinating payments across regions

2. Team 2 — DoorDash (Food Delivery):

- **Workload:** 50K+ concurrent orders during peak hours, 1000+ restaurants, complex routing
- **Characteristics:** Order state machines, route optimization, merchant integration, real-time delivery tracking
- **Critical Requirements:** 99.9% availability, eventual consistency for order status, time-based SLAs (delivery windows)
- **Challenges:** Multi-region orchestration, handling merchant API rate limits, managing delivery driver state

3. Team 3 — Netflix (Streaming):

- **Workload:** 10M+ concurrent streams, 5000+ titles, personalized recommendations
- **Characteristics:** Content encoding/transcoding, CDN coordination, watch history analytics, recommendation engine
- **Critical Requirements:** 99.99% playback availability, sub-3-second startup time, multi-region edge caching
- **Challenges:** Managing massive data transfers, personalization at scale, A/B testing infrastructure

4. Team 4 — Airbnb (Short-Term Rentals):

- **Workload:** 10M+ listings, 1M+ concurrent searches, complex availability calendar management
- **Characteristics:** Search ranking, calendar synchronization, review aggregation, dynamic pricing
- **Critical Requirements:** 99.9% search availability, sub-500ms response times, strong consistency for availability calendars
- **Challenges:** Distributed transaction coordination (booking + payment + calendar), managing calendar race conditions

5. Team 5 — Stripe (Payment Processing):

- **Workload:** 100K+ transactions/second globally, real-time fraud detection, settlement reconciliation
- **Characteristics:** Transaction idempotency, audit trails, PCI compliance, web-hook delivery, reconciliation
- **Critical Requirements:** 99.99% uptime, zero data loss, strict consistency for financial records, encrypted state
- **Challenges:** Handling eventual consistency in distributed ledgers, managing global payment network coordination, fraud ML models

6. Team 6 — Spotify (Music Streaming):

- **Workload:** 500M+ users, 100M+ tracks, 1M+ concurrent playbacks
- **Characteristics:** Personalized recommendations, collaborative filtering, playlist synchronization, offline mode
- **Critical Requirements:** 99.9% playback availability, personalization within 500ms, real-time social features
- **Challenges:** Recommendation ML model serving at scale, managing playlist consistency across devices, A/B testing framework

Four-Delivery Milestone Schedule

- **Delivery 1 (Weeks 1–3):** Requirements document, workload model, non-functional requirements specification
- **Delivery 2 (Weeks 4–7):** Software architecture, service definitions, data architecture, communication patterns
- **Delivery 3 (Weeks 8–11):** Cloud architecture diagram, capacity planning, cost analysis, growth scenario (20× scaling)
- **Delivery 4 (Weeks 12–16):** LocalStack prototype, resilience analysis, final architecture document, cost breakdown

Expected Outputs

Each team will produce:

1. **Requirements & Workload Specification** — Business problem, actors, functional/non-functional requirements, workload model
2. **Software Architecture Document** — System context, service definitions, data schema, communication patterns, ADRs
3. **Cloud Architecture & Capacity Plan** — Infrastructure diagram, compute/network/storage decisions, cost estimation
4. **Resilience Analysis Report** — Failure scenarios (instance failure, AZ outage, database failure), recovery strategies
5. **Final Integrated Project** — LocalStack vertical slice, working dashboards/APIs, comprehensive final report